“Detecting and Mitigating Malware Using Wazuh SIEM and VirusTotal Integration”

Documented By: Kamran Bader

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https://www.linkedin.com/in/kamran-bader-5a8590286/

Executive Summary

This lab demonstrates how to configure Wazuh SIEM to detect and mitigate malware threats by integrating with VirusTotal. The process involves installing Python and PyInstaller, creating an automated malware removal script, and configuring Wazuh to leverage VirusTotal's API for real-time malware detection. A controlled test is performed by downloading and executing a known malicious file after disabling Windows Defender. Wazuh successfully detects the malicious activity and reports it in the dashboard, showcasing its ability to automate malware analysis, detection, and response.

Furthermore, the integration highlights the value of external threat intelligence in strengthening endpoint security. It proves that automated workflows can significantly reduce manual intervention during incident handling. Overall, this approach enhances organizational readiness against advanced malware attacks.

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Pre-Lab Requirements

Before starting the lab, ensure the following prerequisites are met:

1. **Windows VM** with administrator access.
2. **Ubuntu VM (Wazuh server)** with Wazuh all-in-one installed.
3. **Python installed** with “Add to PATH” enabled for all users.
4. **PyInstaller package** installed.
5. **Wazuh Agent** installed on the Windows VM.
6. Access to **VirusTotal API key** (requires an account).
7. Administrative rights to edit Wazuh configuration files.

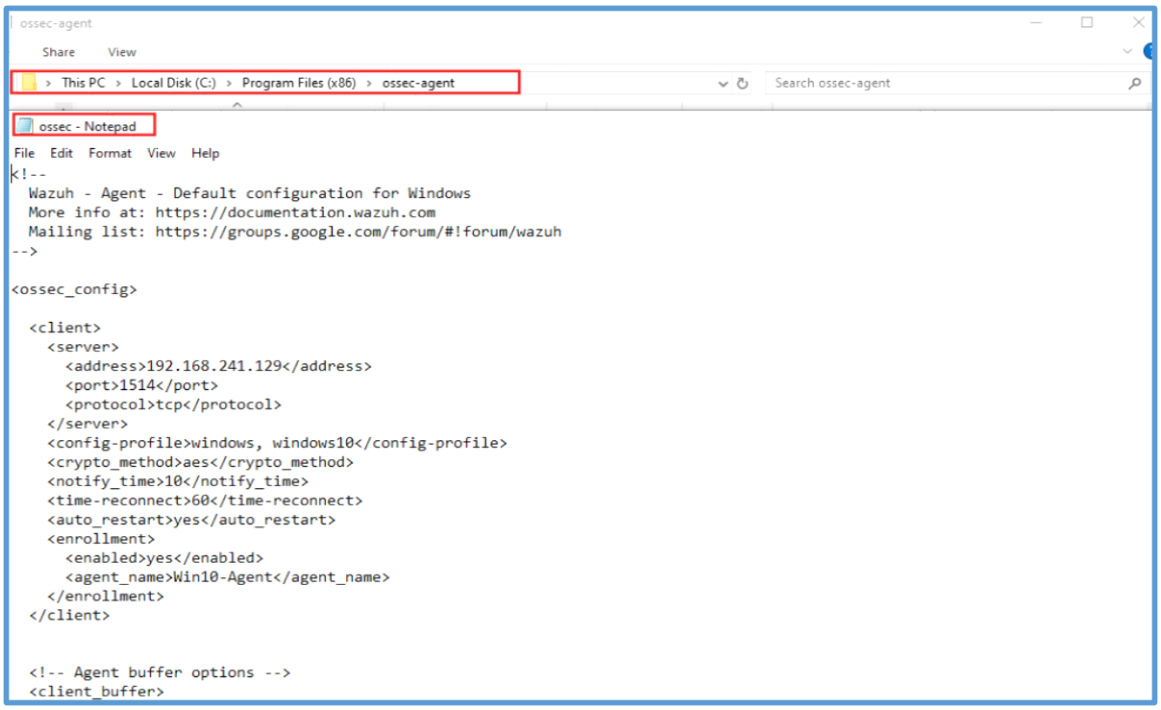
Objective

The objective of this lab is to:

- Demonstrate how to integrate Wazuh SIEM with VirusTotal for malware detection.
- Automate malware removal using custom PowerShell/Python scripts.
- Test Wazuh’s ability to log, detect, and respond to malicious files in real time.
- Validate the effectiveness of Wazuh as an endpoint malware analysis solution.

Step 1: Configure Real-time Monitoring

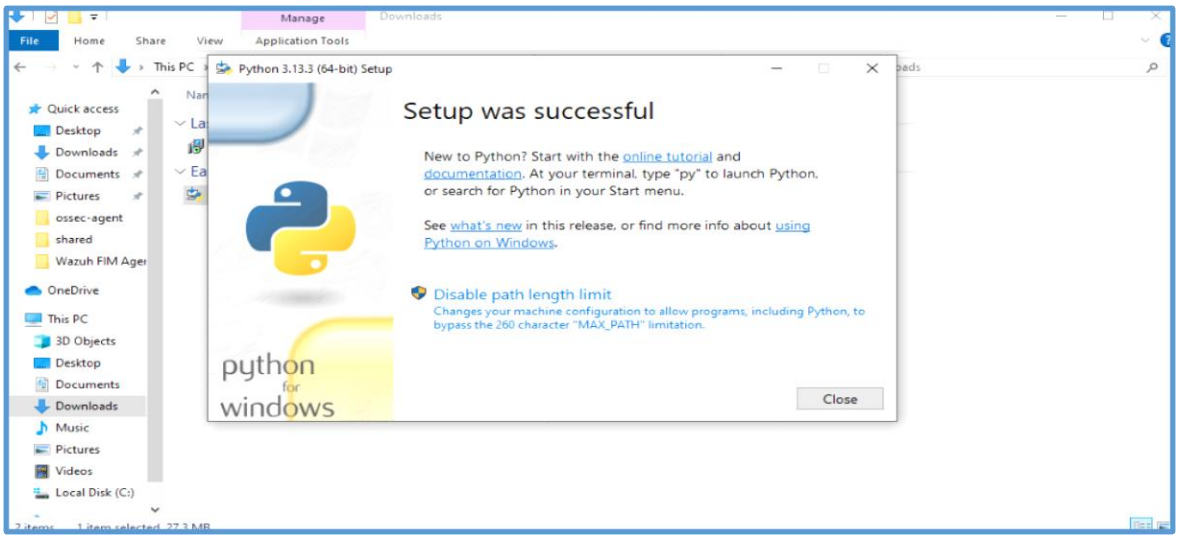
- Add monitoring for downloads directory in ossec.conf:



- <directories realtime="yes">C:\User\kamra\Downloads</directories>
- This ensures that any file added to the Downloads folder is automatically scanned.

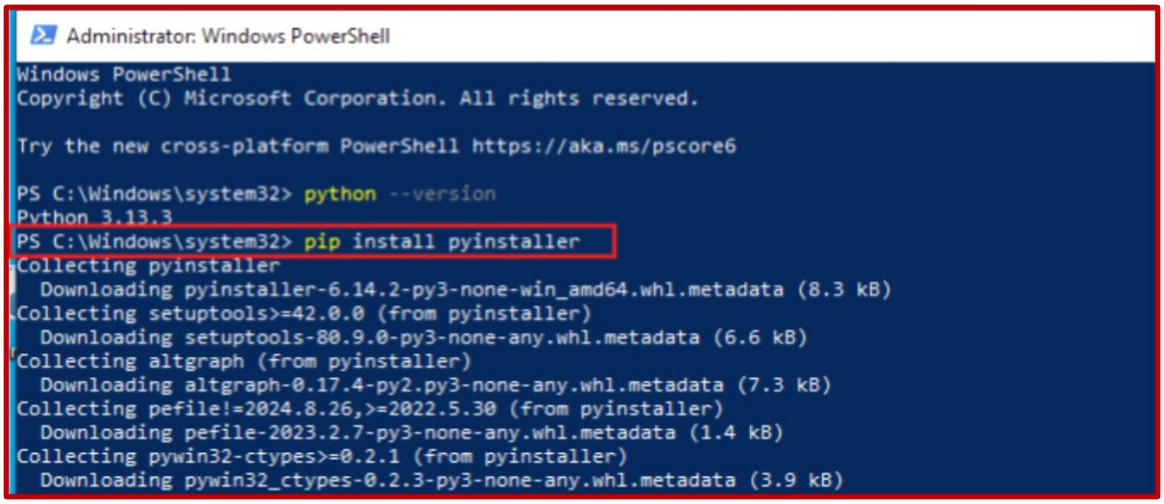
Step 2: Install Python

- Run the Python installer.
- Choose **custom installation**.
- Enable “Install for all users” and “Add to PATH”.



Step 3: Install PyInstaller

- Open PowerShell as Administrator.
- Run:
- pip install pyinstaller



- Verify the installation by checking the PyInstaller version.

```
   ---------------------------------------- 1.4/1.4 MB 238.2 kB/s eta 0:00:00
Downloading packaging-25.0-py3-none-any.whl (66 kB)
Downloading pefile-2023.2.7-py3-none-any.whl (71 kB)
Downloading pyinstaller_hooks_contrib-2025.8-py3-none-any.whl (442 kB)
Downloading pywin32_ctypes-0.2.3-py3-none-any.whl (30 kB)
Downloading setuptools-80.9.0-py3-none-any.whl (1.2 MB)
   ---------------------------------------- 1.2/1.2 MB 391.5 kB/s eta 0:00:00
Downloading altgraph-0.17.4-py2.py3-none-any.whl (21 kB)
Installing collected packages: altgraph, setuptools, pywin32-ctypes, pefile, packaging
taller
Successfully installed altgraph-0.17.4 packaging-25.0 pefile-2023.2.7 pyinstaller-6.14
8 pywin32-ctypes-0.2.3 setuptools-80.9.0

[notice] A new release of pip is available: 25.0.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Windows\system32>
```

Step 4: Create Malware Removal Script

- Download the reference script from Wazuh documentation:
  [Wazuh Remove Threat Script]

```
remove-threat.py - Notepad
File  Edit  Format  View  Help
# Copyright (C) 2015-2025, Wazuh Inc.
# All rights reserved.

import os
import sys
import json
import datetime
import stat
import tempfile
import pathlib

if os.name == 'nt':
    LOG_FILE = "C:\\Program Files (x86)\\ossec-agent\\active-response\\active-responses.log"
else:
    LOG_FILE = "/var/ossec/logs/active-responses.log"

ADD_COMMAND = 0
DELETE_COMMAND = 1
CONTINUE_COMMAND = 2
ABORT_COMMAND = 3

OS_SUCCESS = 0
OS_INVALID = -1

class message:
    def __init__(self):
        self.alert = ""
        self.command = 0
```

- Save it as remove-threat.py on Desktop (ensure “Save as type: All files”).

Step 5: Convert Script to Executable

- In PowerShell, navigate to Desktop.

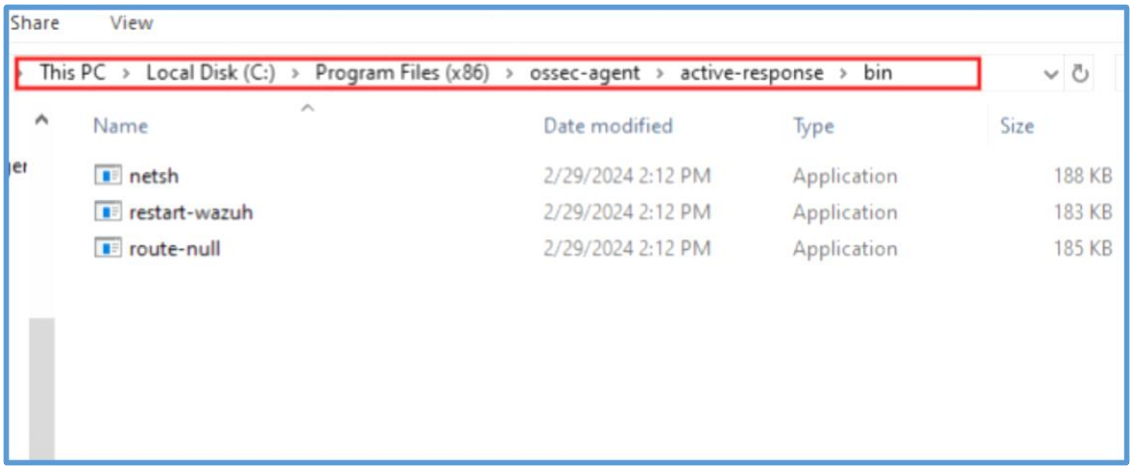
```
Windows PowerShell
PS C:\Users\kamran\Desktop> ls


    Directory: C:\Users\kamran\Desktop


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          8/2/2025  11:05 AM                Wazuh FIM Agent
-a----          6/2/2025  11:22 PM       28640016 python-3.13.3-amd64.exe
-a----          8/2/2025  12:14 PM           4650 remove-threat.py
```

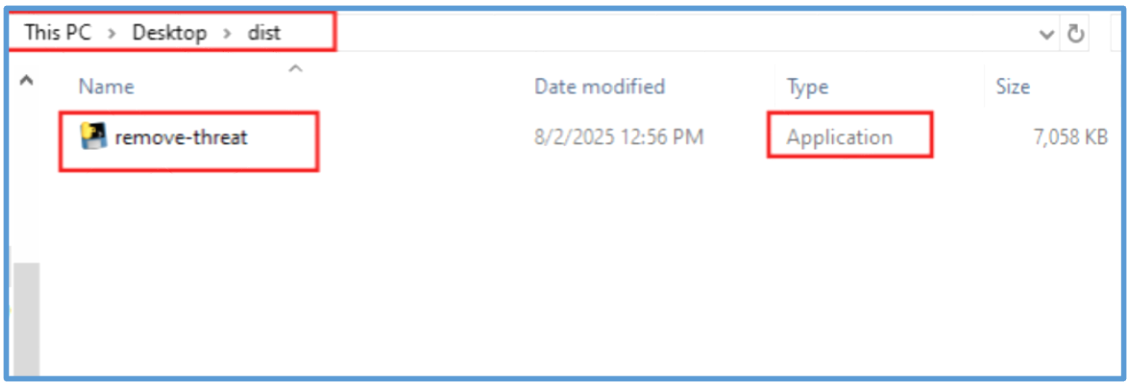
- Run:
- pyinstaller -F remove-threat.py
- This creates a standalone executable.

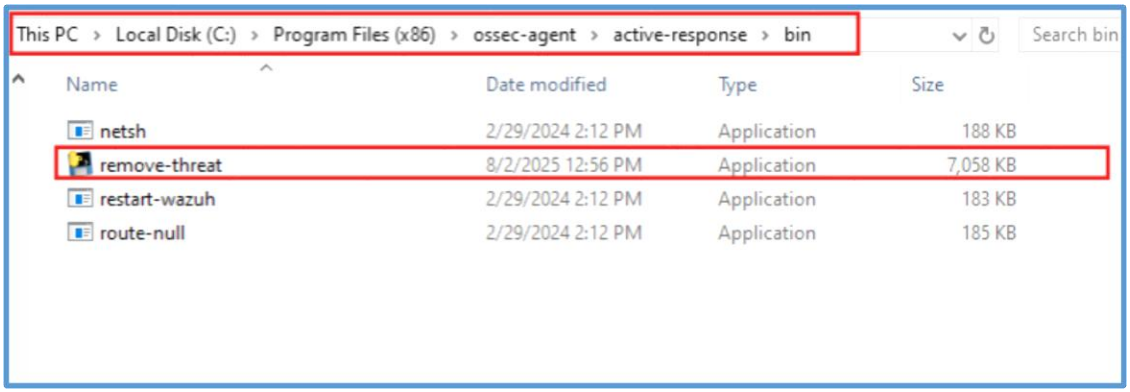
```
PS C:\Users\kamran\Desktop> pyinstaller -F remove-threat.py
1027 INFO: PyInstaller: 6.14.2, contrib hooks: 2025.8
1030 INFO: Python: 3.13.3
1130 INFO: Platform: Windows-10-10.0.19045-SP0
1130 INFO: Python environment: C:\Program Files\Python313
1134 INFO: wrote C:\Users\kamran\Desktop\remove-threat.spec
1191 INFO: Module search paths (PYTHONPATH):
['C:\\Program Files\\Python313\\Scripts\\pyinstaller.exe',
 'C:\\Program Files\\Python313\\python313.zip',
 'C:\\Program Files\\Python313\\DLLs',
 'C:\\Program Files\\Python313\\Lib',
 'C:\\Program Files\\Python313',
 'C:\\Program Files\\Python313\\Lib\\site-packages',
 'C:\\Program Files\\Python313\\Lib\\site-packages\\setuptools\\_vendor',
 'C:\\Users\\kamran\\Desktop']
3158 INFO: checking Analysis
3159 INFO: Building Analysis because Analysis-00.toc is non existent
3160 INFO: Running Analysis Analysis-00.toc
3161 INFO: Target bytecode optimization level: 0
```

Step 6: Move Script to Wazuh Active-Response Folder

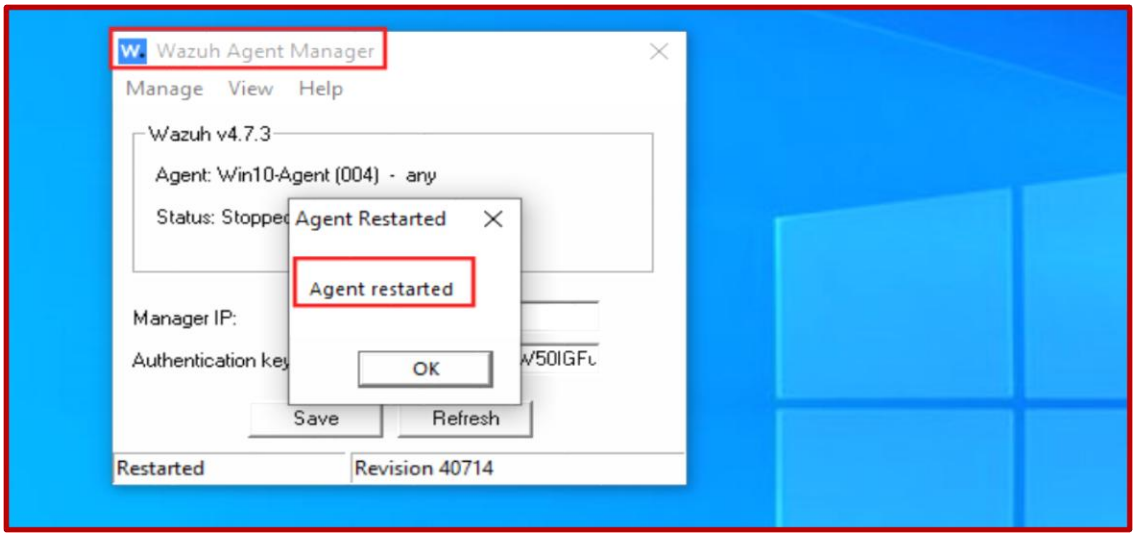
- Copy the generated file into:
- C:\Program Files (x86)\ossec-agent\active-response\bin





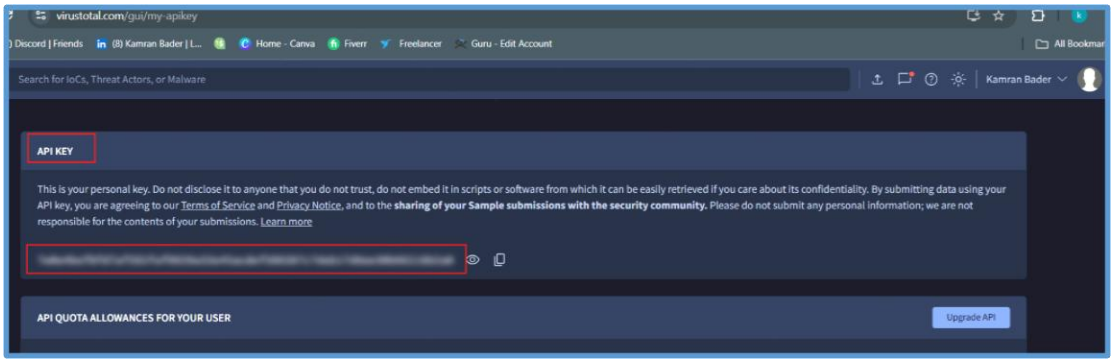


- Restart the **Wazuh Agent** for changes to take effect.



Step 7: Configure VirusTotal Integration

- Create a VirusTotal account.
- Obtain the API key.

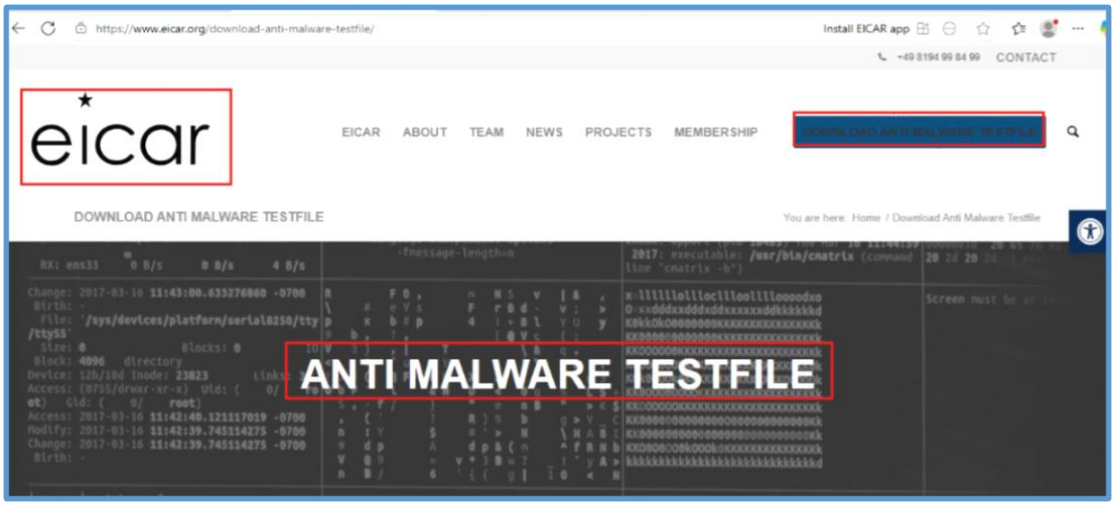


- On the Wazuh server, add the API key into ossec.conf.
- Restart Wazuh Manager.

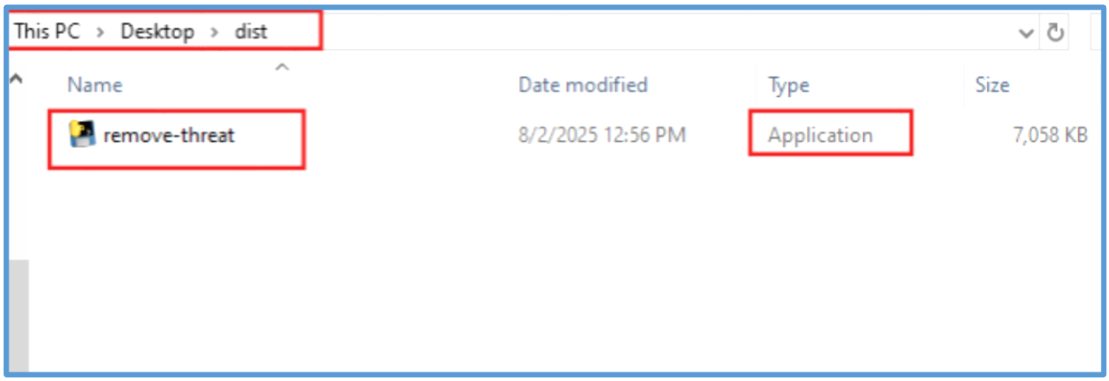
```
GNU nano 7.2                                                    /var/ossec/etc/ossec.conf
    <location>journald</location>
  </localfile>

  <localfile>
    <log_format>apache</log_format>
    <location>/var/log/apache2/error.log</location>
  </localfile>

  <localfile>
    <log_format>apache</log_format>
    <location>/var/log/apache2/access.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

  <integration>
    <name>virustotal</name>
    <api_key>[redacted]33e45acdef508387c7deb17d6ee30b66216b2a0</api_key>
    <group>syscheck</group>
    <alert_format>json</alert_format>
  </integration>

</ossec_config>
```

Step 8: Simulate Malware Infection

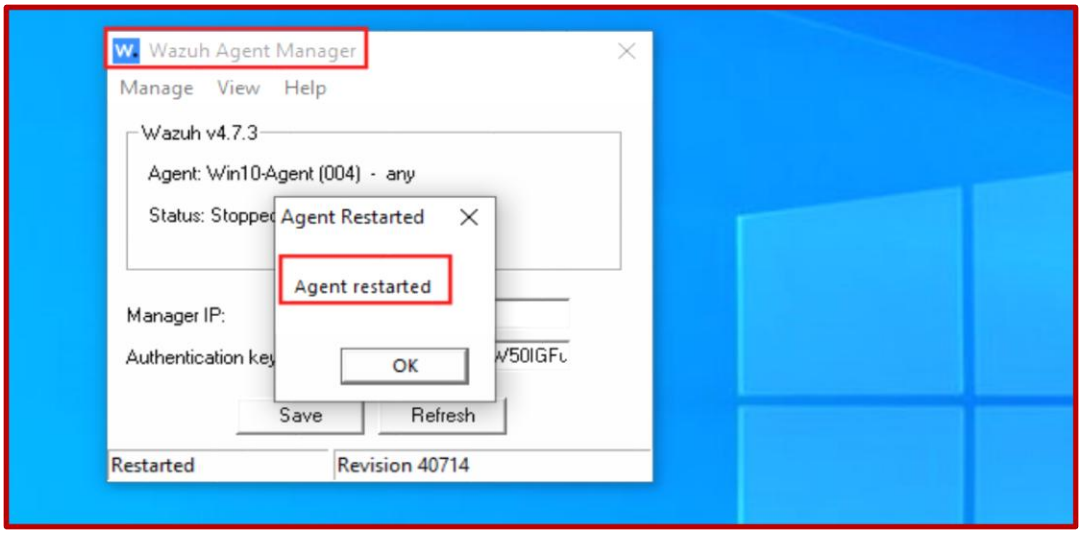
- Disable Windows Defender on the Windows VM.
- Download a **test malware sample** from a known source.



- Copy the malware into the Remove Threat folder.

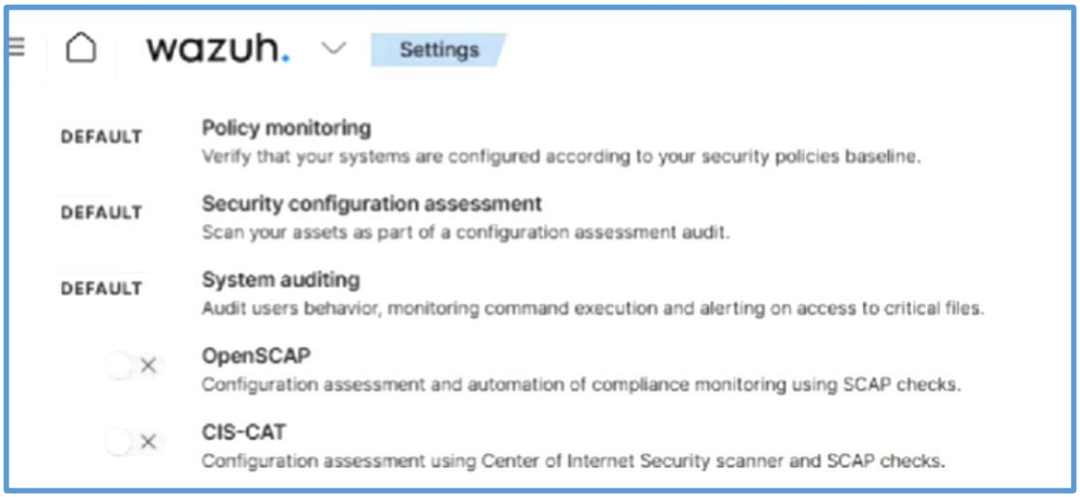


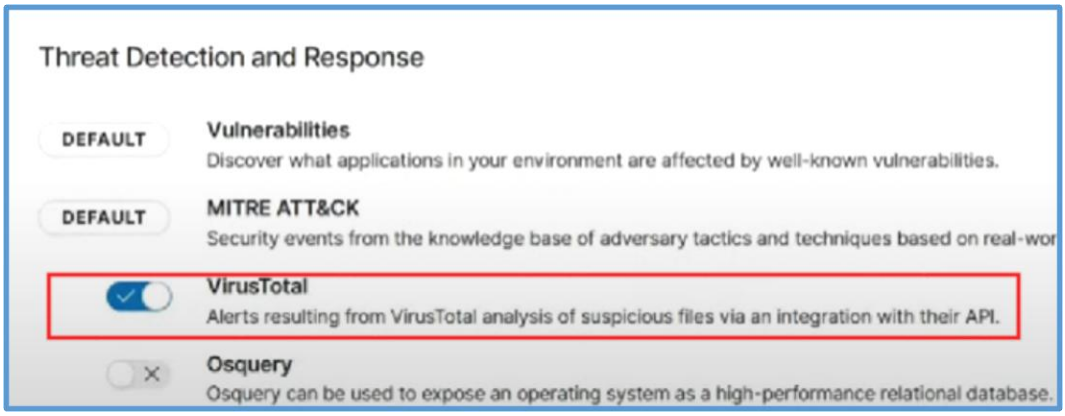
- Extract and execute the file (ignore warning prompt).
- Restart Wazuh Agent.



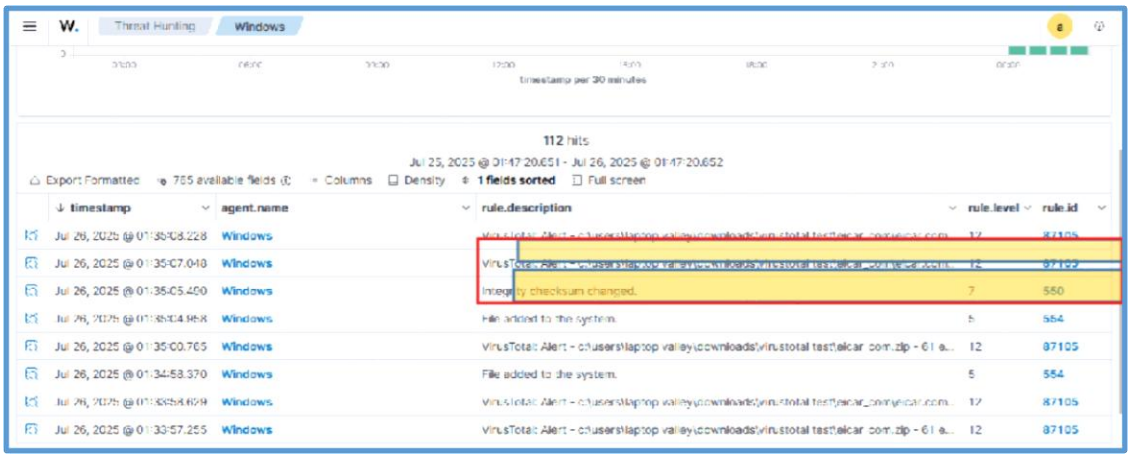
Step 9: Analyze Logs in Wazuh Dashboard

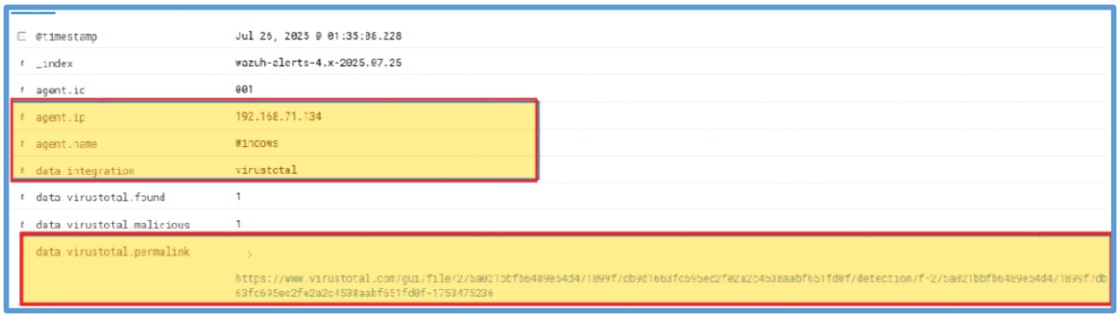
- Open the Wazuh Dashboard.
- Navigate to **Settings → Modules → VirusTotal Module** and enable it.





- Observe the VirusTotal alerts triggered by the malicious file.
- Review detection logs for malware classification details.





Conclusion

This lab successfully showcased the integration of Wazuh SIEM with VirusTotal to automate malware detection and response. By creating an active response script and monitoring directories in real time, Wazuh effectively identified and reported malicious files. The lab demonstrated how SIEM platforms, when combined with third-party threat intelligence (like VirusTotal), can serve as powerful tools for automated malware analysis and mitigation.